Before you solder

V1: keys + knob + display + planned RGB / V2: keys + knob only

Important: Both firmware images and the assembled hardware are untested. V1 below is the **intended source-firmware wiring**; the supplied Full Deck UF2 is unchanged and its pinout, display and LED behavior are **not verified**. Do not assume it matches this diagram. V2 uses the confirmed direct-wired map, but still needs a physical test.

SAME NINE SWITCHES AND ENCODER ON BOTH VERSIONS

Board: **Waveshare RP2040-Zero**. Use the silkscreen GPIO names first. Header numbers below are a cross-check, not a substitute for checking your actual board. These are independent switches, not a key matrix.

Function	GPIO	P1 pin	GND?
K0	GP0	23	Yes
K1	GP1	22	Yes
K2	GP2	21	Yes
K3	GP3	20	Yes
K4	GP4	19	Yes
K5	GP5	18	Yes
K6	GP6	17	Yes
K7	GP11	12	Yes
K8	GP12	11	Yes
Encoder A	GP7	16	COM
Encoder B	GP8	15	COM
Encoder SW	GP9	14	SW2

TOP VIEW OF KEYS

K0 GP0	**K1** GP1	**K2** GP2
K3 GP3	**K4** GP4	**K5** GP5
K6 GP6	**K7** GP11	**K8** GP12

Position: K0–K8 read left to right, then top to bottom. The knob is separate from these nine keys.

P1 orientation: USB-C up, component side toward you. Waveshare numbering starts at 5V = pin 1, then runs counter-clockwise. Verify every label against the board.

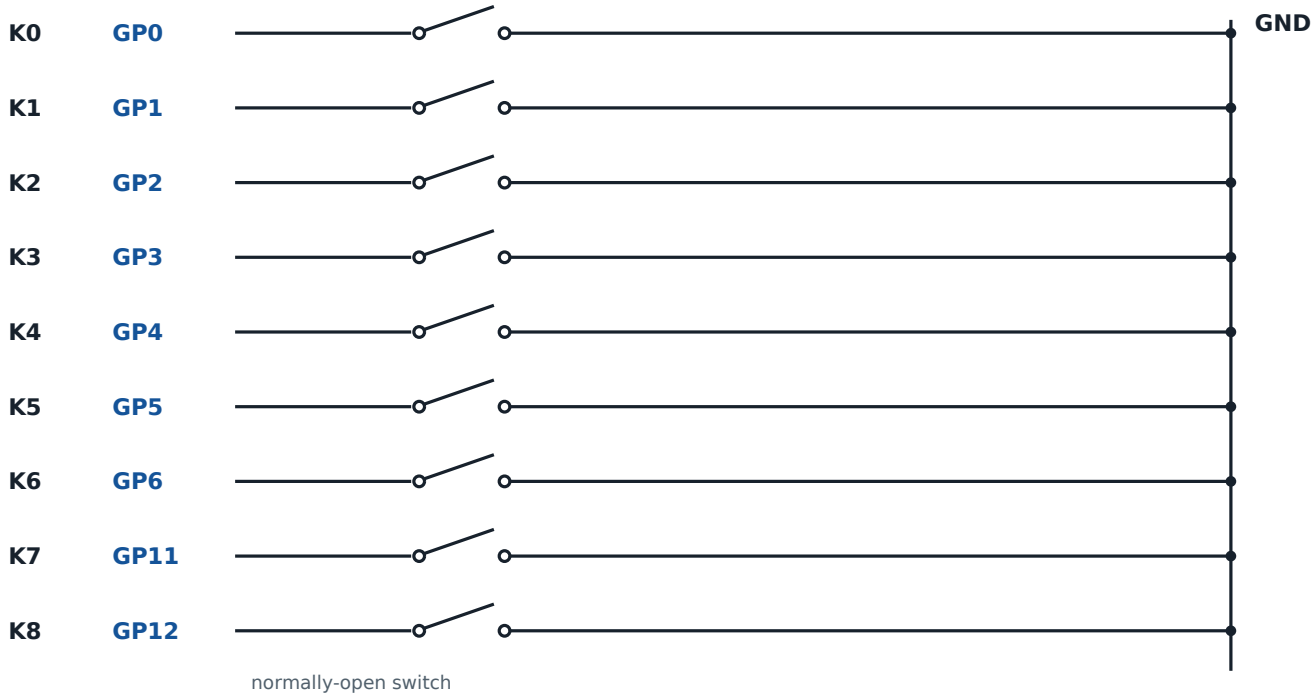
Power rule: unplug USB and every external power supply before soldering or continuity tests. Each key connects one GPIO to GND **only while pressed**; do not connect key contacts to 3V3 or 5V. GPIO inputs use firmware pull-ups. Do not put 5 V onto any RP2040 GPIO.

V2 complete wiring

Keys & Knob only • no OLED and no external RGB wiring

Each row is one normally-open mechanical key. The right-hand vertical line represents **one common GND net**: wire it as a ground daisy chain or insulated distribution point. Do not join the GPIO sides together.

NINE KEYS / VIEW FROM THE SOLDER SIDE, ELECTRICALLY



ENCODER / IDENTIFY CONTACTS BEFORE SOLDERING

ROTATION CONTACTS

```
A   ->  GP7 (P1-16)
B   ->  GP8 (P1-15)
COM ->  GND (P1-2)
```

PUSH SWITCH

```
SW1 -> GP9 (P1-14)
SW2 -> GND (P1-2)
```

Encoder contact order varies: use its datasheet or a meter, not a generic pin picture. Rotation COM and one push-switch contact both go to the common GND net. The metal case is not automatically COM. Swapping A/B reverses direction.

V2 complete: Connect the common GND net to the board GND (P1-2). Leave GP10, GP28 and GP29 unconnected unless independently needed. No key diodes, external pull-ups, OLED or LED supply are required for this version.

The Keys & Knob UF2 is source-built for this map but has not been physically boot-tested. See page 4 for unpowered checks and safe first power-up.

V1 display + planned RGB

Start with all V2 key and encoder wiring on page 2, then add the circuits below

V1 is not solder-ready in full: display/LED part details, LED supply and the bundled Full Deck UF2 compatibility are not established. The LED section is a **proposed topology**, not a complete approved power design.

OLED / SSD1306, 128 X 32, I2C ADDRESS 0X3C

OLED module label	Connect to	Board P1
VCC (3.3 V compatible)	3V3	3
GND	GND	2
SDA	GP28	5
SCL	GP29	4

If the module lacks I2C pull-ups, fit **one approximately 4.7 kΩ resistor from SDA to 3V3 and one from SCL to 3V3**. First check whether the module already has pull-ups and that they pull up to **3V3**. Do not use 5 V pull-ups or assume the module pin order.

RGB / PROPOSED 5 V ADDRESSABLE RGB SIGNAL PATH

GP10 (P1-13) —— **A** — **74AHCT1G125** (5 V buffer) **Y** —— **330 ohm** —— **LED0 DIN**

```
LED0 DOUT -> LED1 DIN; each DOUT -> next DIN
```

Every LED VDD joins the **designed LED 5 V rail**; every LED GND joins shared GND. Follow each pixel's DIN/DOUT arrows. Breakout points and wire sizes depend on the actual strip.

Buffer: VCC → LED 5 V; GND → shared GND; /OE → GND for always-enabled example; A → GP10; Y → 330 Ω → LED0 DIN. Candidate 10 kΩ from A to GND; 100 nF across buffer VCC/GND and 10–47 µF across LED 5 V/GND near LED0 (observe polarity).

Typical TI DBV/DCK/DRL package pins: /OE=1, A=2, GND=3, Y=4, VCC=5. Verify the **exact IC and package datasheet** and the pin-1 mark before soldering; other packages may differ. Use AHCT, not AHC, for this proposed level-shifter.

STOP: The 5 V source, strip type/current, protection, power isolation and power sequencing are not selected. Leave LED power and data disconnected until designed and verified. Never tie external 5 V directly to USB/board 5 V. A common GND is required after an isolated supply design is approved. Do not drive an unpowered buffer/strip.

Checks before power

A meter and a staged test are essential; these diagrams are not a measured circuit

1 / BEFORE SOLDERING

Confirm the exact RP2040-Zero silkscreen, encoder datasheet, switch terminals, OLED voltage/address and, for V1, LED part and supply design. Plan strain relief and insulation; trim wire ends and avoid solder bridges. Do not count on a generic encoder drawing or a module’s connector order.

2 / UNPOWERED MULTIMETER CHECKS

Disconnect USB and all supplies. Check **3V3 to GND** and **5V to GND** for a persistent short; capacitors can briefly beep while charging. For each K0–K8, check its GPIO to GND: open when released, closed only while pressed. Check that separate GPIOs are not shorted together. Check encoder A/B to COM while turning and SW1/SW2 while pressing. Look for bridges and exposed conductors.

3 / POWER AND TEST THE BASIC PAD

Leave LED power and data disconnected. With the circuit inspected and no persistent short, connect USB data to a suitable computer or protected supply. Verify the board enumerates; use the app’s Tester and safe bindings. Physical actions can execute while testing. Serial event capability does not mean event reporting is enabled, and no event alone does not prove a fault.

4 / V1 DISPLAY, THEN RGB ON A SUITABLE BENCH

With power disconnected, fit the confirmed 3.3 V OLED and recheck each net. Test with firmware actually verified for this display and map. For RGB, select a compatible individually addressable strip and calculate/measure worst-case current, including the board and display. Design source isolation and safe startup/shutdown; use a current-limited bench supply and start with one pixel only after the circuit and firmware are verified.

RGB firmware warning: The default firmware source disables external LED data until a deliberate bench-enabled build; settings in the app do not enable output. The supplied Full Deck UF2 is an unchanged, unverified image: its source, pin map and LED behavior are unknown. A low GP10 after initialization is not a power switch and does not guarantee dark LEDs at reset or clear a previously latched color.

PROJECT REFERENCES

Pin map: **firmware/include/board.h**; detailed electrical notes: **docs/HARDWARE.md**; image provenance: **electron/firmware/README.md**. GPIO/header names have been checked against project documentation, not measured on your specific board. Actual component datasheets and a real bench test remain necessary.

Illustrated wiring views follow on pages 5 and 6.

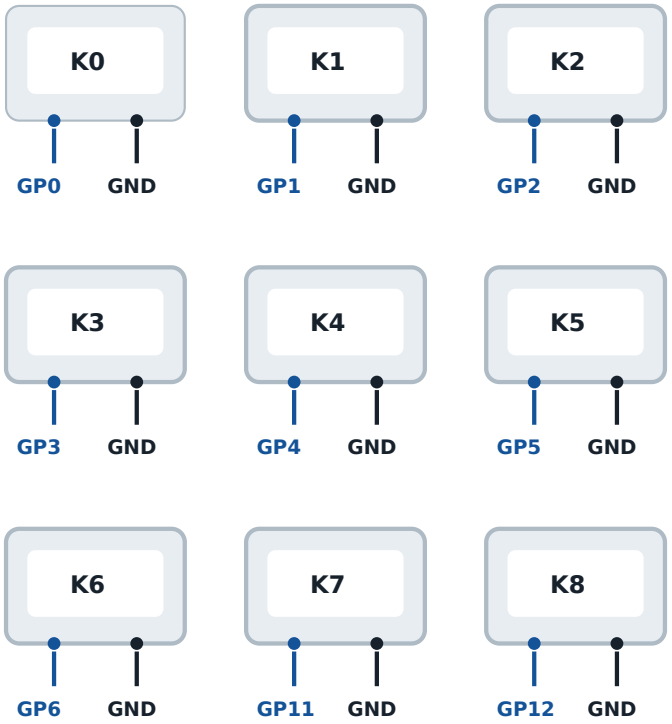
V2 illustrated wiring

Nine switches + encoder / logical connections, not a physical pinout

Read the labels, not the positions: component drawings and board pads below are symbolic. Actual terminal order varies. Every label with the same name is the **same electrical connection**. Match GP labels to the board silkscreen.

SWITCH LAYOUT / K0 TO K8 AS SEEN FROM ABOVE

Each switch has two electrical terminals; drawings are not its solder-side pinout.



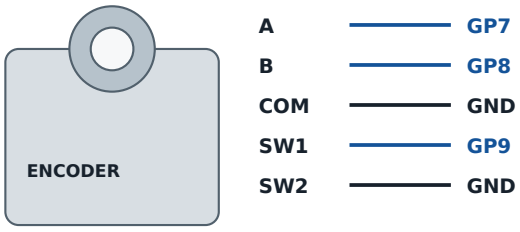
RP2040-ZERO

Functional connection list



Net	P1	
GP0	23	
GP1	22	
GP2	21	
GP3	20	
GP4	19	
GP5	18	K0–K8
GP6	17	
GP11	12	
GP12	11	
GP7	16	
GP8	15	
GP9	14	Encoder
GND	2	
		Common

ENCODER / SYMBOLIC CONTACTS



Find the real terminals using the encoder datasheet or a multimeter. Rotary COM and one switch contact both go to GND. Do not assume the case or the pictured positions are the real contacts.

Connect all GND labels together and to RP2040-Zero GND (P1-2). Each blue GPIO goes to only its named key or encoder contact. No OLED, LED strip or LED power is used on V2. Complete electrical circuit and checks: pages 2 and 4.

V1 illustrated additions

First wire all keys + encoder exactly as shown on pages 2 and 5

V1 firmware and RGB are not yet verified. The Full Deck UF2 may not match this source pin map. The LED drawing is a **proposed concept**; leave its power and data disconnected until parts, supply and firmware are checked.

OLED / MATCH LABELS ON YOUR ACTUAL MODULE

Connector order is illustrative only. Read your module labels before wiring.

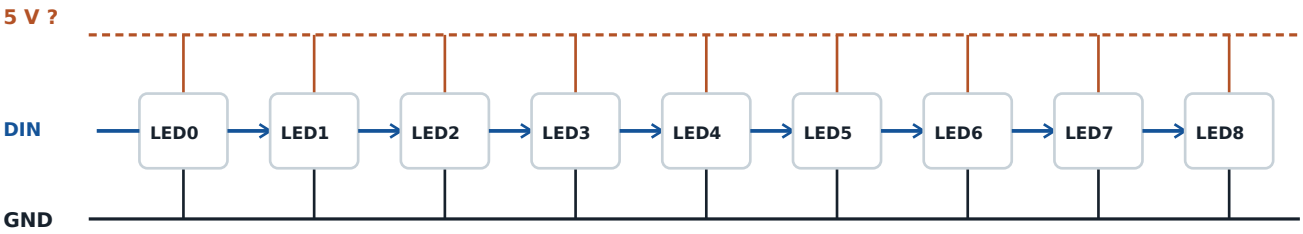


Use a **3.3 V compatible** 128×32 SSD1306 I2C module at 0x3C. SDA/SCL need pull-ups to **3V3** (about 4.7 kΩ if not already fitted). Verify VCC and I2C pull-up voltage and module connector order.

RGB / CONCEPTUAL CHAIN; POWER SOURCE NOT DESIGNED



Buffer power, /OE, capacitors and safety details: page 3.



Blue = data, DOUT to next DIN / brown dashed = proposed 5 V / black = GND

No LED supply is selected or approved. Each pixel's VDD and GND must follow the purchased part's datasheet; arrows show data direction. Do not assume cut points, 5 V current, board/USB capacity or 3.3 V data compatibility. Do not directly join an external 5 V source to USB 5 V. Common GND, isolation and power sequencing need a verified design. The drawings do not establish physical pixel pin orientation.